

AUTOMATING JOB APPLICATIONS FROM SCRAPING TO COVER LETTER

: n8n + Ampify + ChatGPT 4.0 mini

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Motivation

- Why job hunting is repetitive and time-consuming
- Challenges: reading job details, evaluating fit, and writing cover letters
- Need for automation in the modern job-seeking process





Objectives

- Scrape live job data from Indeed
- Rate job relevance to user preferences
- Generate a tailored cover letter
- Fully automate the process using n8n

Tools and Tech Stack

Tool	Purpose
n8n	Workflow orchestration (low-code)
Ampify	Web scraping (job listing extraction)
ChatGPT 4.0 mini	AI for understanding, rating, and generating cover letters
HTTP/Webhook nodes	Trigger & API calls
Set/Function nodes	Data formatting



Step 1 : SET UP A TRIGGER

1. First step is to set up a trigger.
2. A trigger runs the whole automation process on a set time.
3. For this project I set the trigger to everyday, at noon.

 Schedule Trigger1

Parameters

Settings

Docs 

This workflow will run on the schedule you define here once you **activate** it.

For testing, you can also trigger it manually: by going back to the canvas and clicking 'execute workflow'

Trigger Rules

Trigger Interval

Days

Days Between Triggers

1

Trigger at Hour

Noon

Trigger at Minute

0

Step 2 – Job Scraping using Ampify

- Scrapes jobs from Indeed based on keywords/filters
- Captures job title, company, location, description, salary (if any)
- Uses CSS selectors or HTML pattern matching
- Sends scraped data to n8n via webhook

Mayank
Personal

Search...

K

Get started

1 / 4 steps

Apify Store

Actors

Runs

Saved tasks

Integrations

Schedules

Development

My Actors

Insights

Messaging

Proxy

RAM0 MB / 8 GB

Usage\$0.17 / \$5.00

< All Actors



Indeed jobs scraper [PPR]

\$5.00 / 1,000 Jobs ▾

borderline/indeed-scraper

★★★★★ 5.0 (1)

🔒 50

👤 626

👤 265

Crafted by borderline

Maintained by Community

▶ Start

📄 Create task

API ▾

...

Pay per result - Fast and reliable Indeed Job Scraper! Extract job listings with company details effortlessly with advanced filters, avoid blocking, and enjoy lightning-speed execution. Build your analytics, automate searches, or benchmark salaries—all with seamless outputs. Try it now for free

Input

Information

Runs 4

Builds 8

Integrations 0

Monitoring

Issues 2

Saved tasks 0

Reviews

Manual

JSON

»

Important: Choose Only One Option

- Search Query & Filters OR URLs
- Options cannot be combined—parameters from one option will be ignored if the other is selected.

▾ Option 1: Using search query & filters

Important Usage Notice

- Country and Query are REQUIRED fields when using traditional search parameters to retrieve data from

HTTP REQUEST.

1. On the side is the http request from Ampify actor to our n8n account.
2. In the edit JSON section we can see all the selections we have chosen.
3. Most important selection would be to select max rows to be 10.
4. If we don't set the max rows section, the Actor will scrape over 1000 listing at a time and cost a lot.

Edit JSON

```
1 {  
2     "country": "us",  
3     "fromDays": "3",  
4     "jobType": "fulltime",  
5     "level": "entry_level",  
6     "location": "USA",  
7     "maxRows": 10,  
8     "maxRowsPerUrl": 10,  
9     "postedBy": "employer",  
10    "query": "Software engineer",  
11    "sort": "date",  
12    "urls": [  
13        "https://www.indeed.com/jobs?q=software+engineer",  
14    ]  
15 }
```

Output of HTTP request

1. On the side is the output of the run.
2. As expected we have our 10 jobs along with their description and title.
3. We can pin a single result to make sure our entire workflow works before running all 10 job postings.



```
10 items
[
  {
    "jobKey": "dc027b2b5a99c8f1",
    "scrapingInfo": {
      "page": 1,
      "index": 0
    },
    "title": "Entry-Level Software Developers",
    "jobType": [
      "Full-time"
    ],
    "descriptionHtml": "<div><p><b>Position Overview</b></p>
      <p>\nTruTechnologies is seeking entry-level and early-career software developers to join our small but growing team. You must have a passion for diving deep into real-world
```


CHAT- GPT MODEL 1

1. Now we will put all the information in JSON for to chat gpt and provide it with a system and user prompt.
2. System prompt: You are an intelligent bot capable of pulling out data from job listing sites.
3. User prompt: Providing the JSON site.
4. Our last prompt is an assistance prompt asking to retrieve all the important information out of the large JSON format.

Prompt

You're an intelligent bot capable of pulling out data from a job listing site.

Role

System

Prompt

fx

Here's the job site:
{{ \$json.data }}

Here's the job site: <!DOCTYPE html><!-- Last Published: Sun Jun 29 2025 03:11:56 GMT+...

Role

User

Please output the data in JSON format, using the following structure:

{
 "company_name": "",
 "benefits": "", //401K plan, medical, etc, please separate by commas
 "job_description": "", //please make it 200 characters or less
 "location": ""
}

OUTPUT of MODEL 1

- The output from model 1 provides us with all the needed information.
- If any required information is not provided on the job listing, it will return with 0.

1 item

```
{  
  "message": {  
    "role": "assistant",  
    "content": {  
      "company_name": "TruTechnologies",  
      "benefits": "",  
      "job_description": "TruTechnologies is seeking entry-level software developers who are proactive, eager to learn, and ready to tackle real-world business problems through technology.",  
      "location": "Research Triangle Park, NC"  
    },  
    "refusal": null,  
    "annotations": [  
    ]  
  },  
  "logprobs": null,  
  "finish_reason": "stop"  
}
```

CHAT-GPT model 2

From list ▾ GPT-4O-MINI ▾

Messages

Prompt

You're an intelligent bot rating how closely a job listing is to a candidates skill set, on a score of 5.

Give the following points:

- a) 3 points for skills matching, 1 point for mostly matching
- b) 1 point for it being the right experience level
- c) 1 point for it being a remote position
- d) 1 point if past job experience matches the job role
- e) 1 point if the skills on the resume align with the job
- f) 3 points if I meet the job qualifications posted based on my resume

Role

System ▾

For the second model we set the system with some instructions as seen in the previous slide.

System needs to give us a rating out of 5 to see the chances of us getting alligning with our career.

Then we provide our resume to the model to give us the rating.

Here's the job listing details:

```
{{ $('HTTP Request').item.json.data }}
```

fx

Here is the candidates skill set:

Mavank Singh

Here's the job listing details: <!DOCTYPE html><!-- Last Published: Sun Jun 29 2025 03:11:...

Role

User

OUTPUT

- After running the model. We are given an output score of 5 based on how well it alligns with our resume and requirements.

1 item

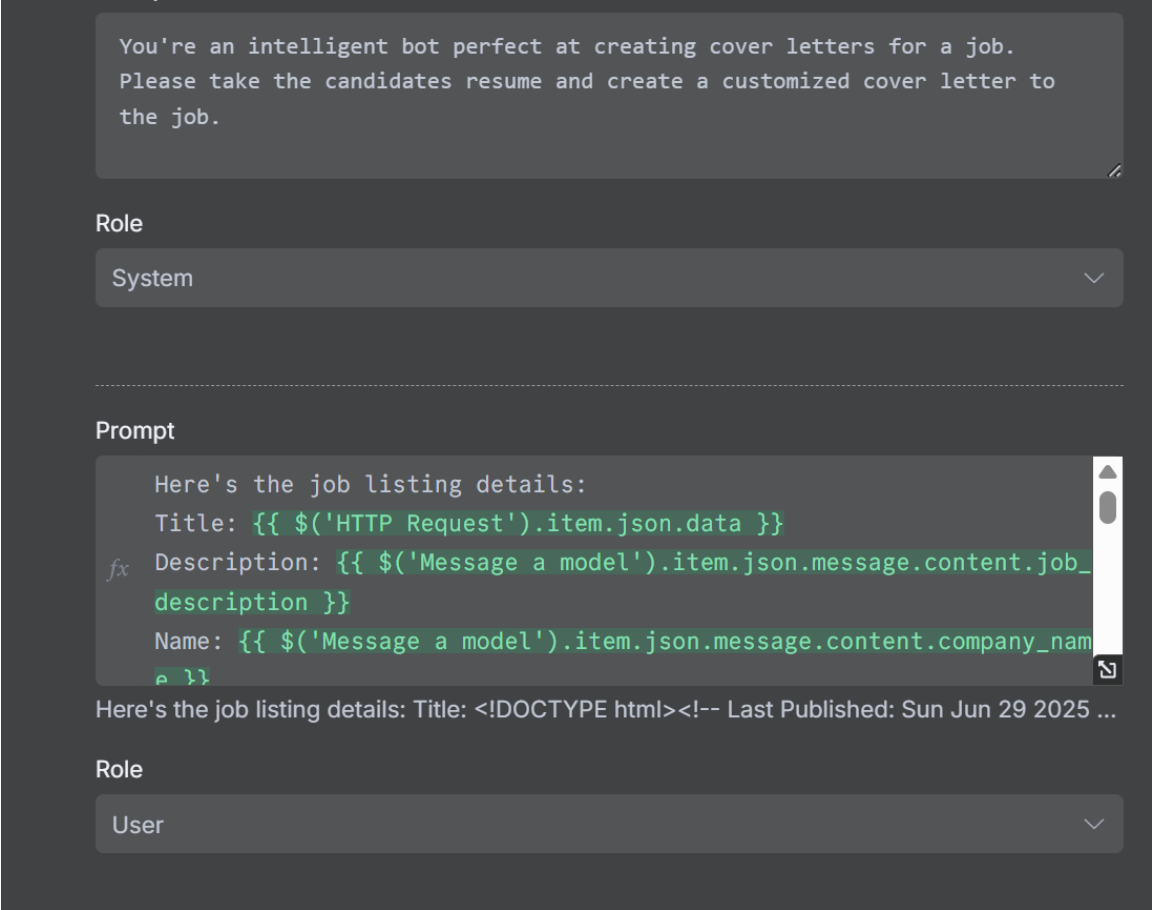
```
"id": "chatcmpl-  
Bq1mpSF2SThN7bSmFeGa4vrzMt4SH",  
"object": "chat.completion",  
"created": 1751739623,  
"model": "gpt-4o-mini-2024-07-18",  
"choices": [  
  {  
    "index": 0,  
    "message": {  
      "role": "assistant",  
      "content": {  
        "score": "5"  
      },  
      "refusal": null,  
      "annotations": [  
      ]  
    },  
    "logprobs": null,  
    "finish_reason": "stop"  
  }  
]
```

CHAT-GPT MODEL 3

For our third model we have our system prompt set as

System Prompt: You're an intelligent bot perfect at creating cover letters for a job.

For our user prompt we provide the model with the job listing and our resume to provide a unique cover letter for all job postings.



The screenshot shows a ChatGPT interface with a dark theme. At the top, a system prompt is displayed in a light gray box: "You're an intelligent bot perfect at creating cover letters for a job. Please take the candidates resume and create a customized cover letter to the job." Below this, the "Role" is set to "System". A dashed line separates the system prompt from the user prompt. The "Prompt" section shows a user prompt with a code editor interface. The code is as follows:

```
Here's the job listing details:  
Title: {{ $('HTTP Request').item.json.data }}  
Description: {{ $('Message a model').item.json.message.content.job_  
description }}  
Name: {{ $('Message a model').item.json.message.content.company_name }}
```

Below the code editor, the rendered HTML is shown: "Here's the job listing details: Title: <!DOCTYPE html><!-- Last Published: Sun Jun 29 2025 ...". At the bottom, the "Role" is set to "User".

OUTPUT

Output provides us with the cover letter for our job.

```
1 item

{
  "created": 1751739624,
  "model": "gpt-4o-mini-2024-07-18",
  "choices": [
    {
      "index": 0,
      "message": {
        "role": "assistant",
        "content": {
          "cover_letter": "Mayank Singh\n814-418-8274\nmayank.goa@gmail.com\n\n[Date]\n\nHiring Manager\nTruTechnologies\nResearch Triangle Park, NC\n\nDear Hiring Manager,\n\nI am writing to express my interest in the Entry-Level Software Developer position at TruTechnologies as
```

GOOGLE SHEET

For our final part of the flow, we provide all the information as an input for our google sheet.

The google sheet will keep all job posting based on unique links so that the same job doesn't not get in the sheet again.

Link (using to match)

`fx {{ $('HTTP Request1').item.json.applyUrl }}`

<https://www.trutechnologies.io/jobs/entry-level-software-developers>

🗑 Title

`fx {{ $('HTTP Request1').item.json.title }}`

Entry-Level Software Developers

🗑 Job Description

`fx {{ $('Message a model').item.json.message.content.job_description }}`

TruTechnologies is seeking entry-level software developers who are proactive, eager to le...

🗑 Date

`fx`

[empty]

🗑 Rating

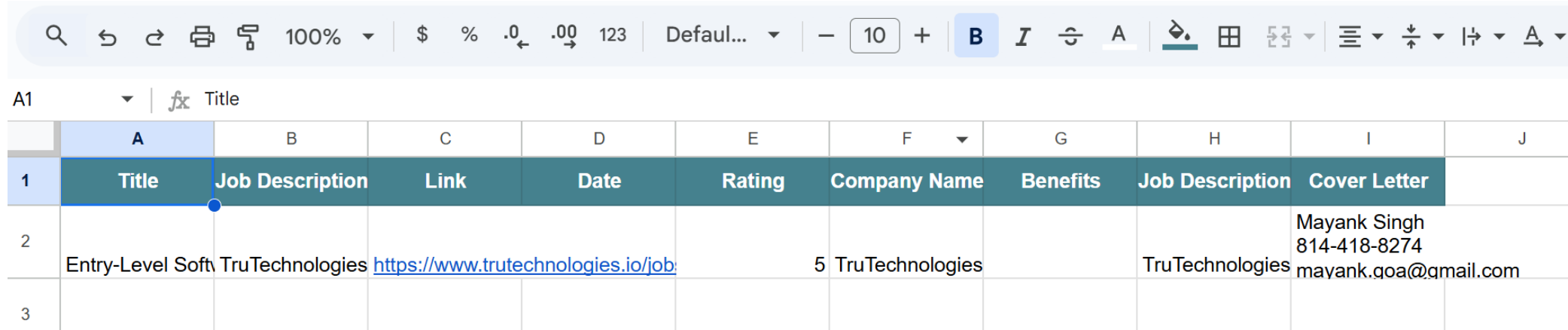
`fx {{ $('Message a model1').item.json.choices[0].message.content.score }}`

5

🗑 Company Name

`fx {{ $('Message a model').item.json.message.content.company_name }}`

Output



The screenshot shows a Google Sheet interface. The toolbar at the top includes search, undo, redo, print, copy, paste, zoom (100%), currency, percentage, decimal, thousandths, text color, background color, border, fill color, text alignment, horizontal alignment, vertical alignment, and link. The sheet has columns A through J and rows 1 through 3. The first row (row 1) contains headers: A (Title), B (Job Description), C (Link), D (Date), E (Rating), F (Company Name), G (Benefits), H (Job Description), I (Cover Letter), and J (empty). The second row (row 2) contains data: A (Entry-Level Soft), B (TruTechnologies), C (<https://www.trutechnologies.io/job/>), D (empty), E (5), F (TruTechnologies), G (empty), H (TruTechnologies), I (Mayank Singh, 814-418-8274, mayank.goa@gmail.com), and J (empty). The third row (row 3) is empty.

	A	B	C	D	E	F	G	H	I	J
1	Title	Job Description	Link	Date	Rating	Company Name	Benefits	Job Description	Cover Letter	
2	Entry-Level Soft	TruTechnologies	https://www.trutechnologies.io/job/		5	TruTechnologies		TruTechnologies	Mayank Singh 814-418-8274 mayank.goa@gmail.com	
3										

- The sheet gets updated everyday at noon with all the important information like.
- Company name, Job description and Link to the job posting.
- 2 really important features are the rating and unique cover letter for all the job postings.

FINAL WORKFLOW

